

# PHYS205 Experiment 5 Lab Report: Stefan-Boltzmann Law

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## 1 Objective

In this experiment, we made relative measurements of the power per unit area emitted from a hot object, namely the Stefan-Boltzmann Lamp, at various temperatures and heat radiation measurements emitted by the palm of your hand to estimate the total power radiated by human body.

## 2 Theory

### Part 1: Stefan-Boltzmann Law

According to Stefan-Boltzmann's law, the energy emitted by a black body per unit area and unit time is proportional to the power "four" of the absolute temperature of the body. Stefan-Boltzmann's law is also valid for a so-called "grey" body whose surface shows a wavelength-independent absorption-coefficient of less than one.

The equation is:

$$H = e\sigma(T^4) \quad (1)$$

where  $\sigma = 5.6703 \times 10^{-8} W/m^2 K^4$ .

### Part 2: Radiation from Human Body

When the palm of the hand is placed in front of the thermopile sensor opening, all the heat radiation from the area of the palm inside the field of view

of the sensor will be detected by the sensor. Then the following formula can be used in order to estimate the heat radiation from the total surface area of a person.

$$P_{Total} = P_{Measured} \frac{A_{Total}}{A_{Source}} \quad (2)$$

Where  $P_{Total}$  the total power radiated by human body,  $P_{Measured}$  is the measured power by the thermopile,  $A_{Total}$  is the total surface area of the human body,  $A_{Source}$  is the area of the hand inside the field of view of the thermopile. The sensitivity of the thermopile is 28 mV/W. The effective area of the sensor is  $2.25 \times 10^{-4} m^2$ . Total surface area of the human body can be calculated by the following equation:

$$A_{Total}(m^2) = 0.202 \times M^{0.425} \times H^{0.725} \quad (3)$$

Measured power by the sensor can also be expressed as;

$$P_{Measured} = A_{Sensor} e \sigma (T_1^4 - T_2^4) \quad (4)$$

Where  $A_{Sensor}$  is the effective sensor area. Then the below formula also gives the expected total heat radiation from the human body:

$$P_{Total} = A_{Sensor} e \sigma (T_1^4 - T_2^4) \frac{A_{Total}}{A_{Source}} \quad (5)$$

Where the emissivity  $e$  of human skin is 0.98,  $T_1$  is the temperature of your hand and  $T_2$  is the room temperature.

### 3 Procedure

Setup the equipment as shown in Figure 1. The voltage sensor should be connected directly to the binding posts of the Stefan-Boltzmann Lamp. A voltage of up to 8 V DC can be applied to the lamp if the period of supply is limited to a few minutes. Moll-type Thermopile sensor is used to measure the heat radiation. It is an electronic device that converts thermal energy into electrical energy. Before a reading is taken, a proper “zero”-adjustment has to be assured. This is done by taking the lamp together with its slide-mount away from the bench for a few minutes. After the lamp has been put

back onto the bench, the reading can be taken if the thermopile has reached its equilibrium. This takes about one minute. Care must be taken that no background radiation disturbs the measurement.

## Equipment

The following equipment is used in this experiment:

- Science workshop 750 interface
- Voltage Sensor and Ammeter
- Thermopile Sensor
- Stefan-Boltzmann Lamp
- Voltage Amplifier and Temperature Sensor

## Part 1: Stefan-Boltzmann Law

The resistance of the tungsten filament at room temperature (about 300K),  $R_{Ref} = 0.541\Omega$ .

1. Set the gain of the voltage amplifier to 1mV. This setting amplifies the reading of the thermopile voltage by a factor of 1000. And zero the thermopile reading.
2. Set the filament voltage to 8V, then repeat for 7V, 6V, 5V, and 4V.
3. Measure the voltage and current into the filament to calculate the resistance  $R$ .
4. Divide  $R$  by  $R_{Ref}$  to obtain the relative resistance ( $R/R_{Ref}$ ).
5. Determine the temperature of the filament using the provided resistivity table (Table 1) or Equation 6.

$$T = \left(\frac{R - R_{Ref}}{\alpha R_{Ref}}\right) + T_{Ref} \quad (6)$$

Table 1: Temperature and Resistivity for Tungsten

| $R/R_{300K}$ | Temp<br>K | Resistivity<br>$\mu\Omega \text{ cm}$ | $R/R_{300K}$ | Temp<br>K | Resistivity<br>$\mu\Omega \text{ cm}$ | $R/R_{300K}$ | Temp<br>K | Resistivity<br>$\mu\Omega \text{ cm}$ | $R/R_{300K}$ | Temp<br>K | Resistivity<br>$\mu\Omega \text{ cm}$ |
|--------------|-----------|---------------------------------------|--------------|-----------|---------------------------------------|--------------|-----------|---------------------------------------|--------------|-----------|---------------------------------------|
| 1.0          | 300       | 5.65                                  | 5.48         | 1200      | 30.98                                 | 10.63        | 2100      | 60.06                                 | 16.29        | 3000      | 92.04                                 |
| 1.43         | 400       | 8.06                                  | 6.03         | 1300      | 34.08                                 | 11.24        | 2200      | 63.48                                 | 16.95        | 3100      | 95.76                                 |
| 1.87         | 500       | 10.56                                 | 6.58         | 1400      | 37.19                                 | 11.84        | 2300      | 66.91                                 | 17.62        | 3200      | 99.54                                 |
| 2.34         | 600       | 13.23                                 | 7.14         | 1500      | 40.36                                 | 12.46        | 2400      | 70.39                                 | 18.28        | 3300      | 103.3                                 |
| 2.85         | 700       | 16.09                                 | 7.71         | 1600      | 43.55                                 | 13.08        | 2500      | 73.91                                 | 18.97        | 3400      | 107.2                                 |
| 3.36         | 800       | 19.00                                 | 8.28         | 1700      | 46.78                                 | 13.72        | 2600      | 77.49                                 | 19.66        | 3500      | 111.1                                 |
| 3.88         | 900       | 21.94                                 | 8.86         | 1800      | 50.05                                 | 14.34        | 2700      | 81.04                                 | 26.35        | 3600      | 115.0                                 |
| 4.41         | 1000      | 24.93                                 | 9.44         | 1900      | 53.35                                 | 14.99        | 2800      | 84.70                                 |              |           |                                       |
| 4.95         | 1100      | 27.94                                 | 10.03        | 2000      | 56.67                                 | 15.63        | 2900      | 88.33                                 |              |           |                                       |

## Part 2: Radiation from Human Body

1. Set the gain of the voltage amplifier to 1mV. This setting amplifies the reading of the thermopile voltage by a factor of 1000. And zero the thermopile reading.
2. An Aluminum-styrofoam sheet with a circular hole is used to fix the source area.
3. Place the sheet between the thermopile sensor and the palm of your hand.
4. Record the voltage sensor reading and measure the temperature of the palm and room.

## 4 Data and Analysis

### Part 1: Stefan-Boltzmann Law

Where the temperature coefficient of resistance for tungsten,  $\alpha = 4.4 \times 10^{-3}\text{K}^{-1}$

Table 2: Lamp Measurements (Table 2)

| V (V) | I (A) | R ( $\Omega$ ) | $R/R_{Ref}$ | T (K) | $T^4 \text{ (K}^4\text{)}$ | $V_{Sensor} \text{ (V)}$ |
|-------|-------|----------------|-------------|-------|----------------------------|--------------------------|
| 8V    | 1.43  | 5.594          | 10.34       | 1823  | $1.104 \times 10^{13}$     | 6.06                     |
| 7V    | 1.33  | 5.263          | 9.73        | 1564  | $8.03 \times 10^{12}$      | 4.83                     |
| 6V    | 1.24  | 4.839          | 8.94        | 1505  | $5.14 \times 10^{13}$      | 3.65                     |
| 5V    | 1.13  | 4.425          | 8.18        | 1332  | $3.14 \times 10^{12}$      | 2.6                      |
| 4V    | 1.01  | 3.960          | 7.32        | 1136  | $1.67 \times 10^{12}$      | 1.71                     |

In order to find  $T$  and  $T^4$ , I used the Equation 6,

$$T = \left( \frac{\frac{V}{I} - 0.541}{4.4 \times 10^{-3} \cdot 0.541} \right) - 300$$

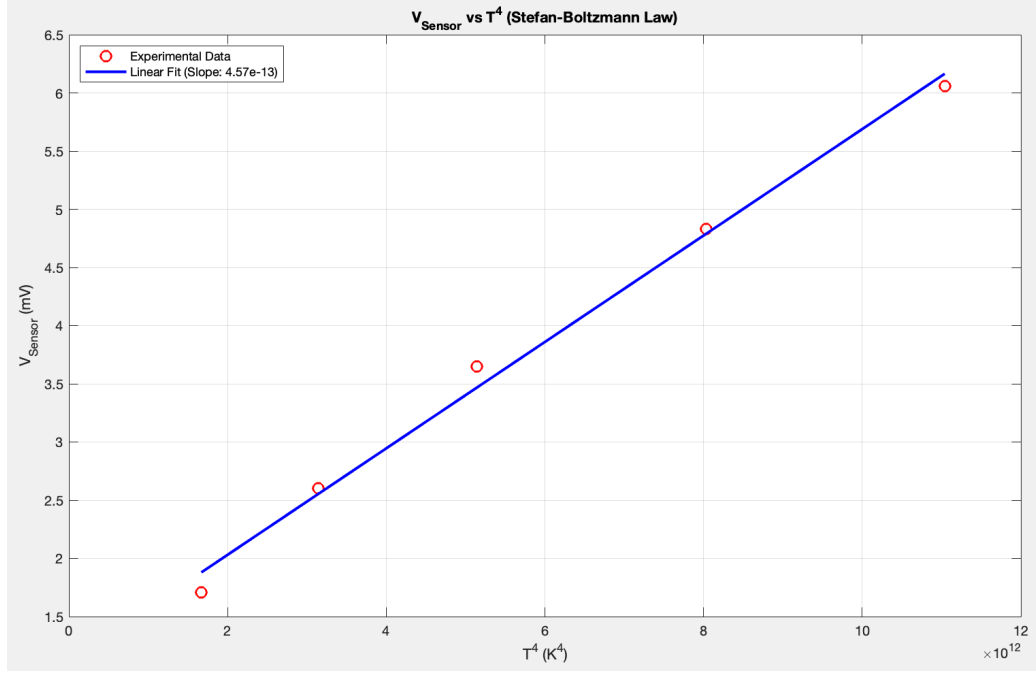


Figure 1:  $V_{Sensor}$  vs  $T^4$

## Part 2: Radiation from Human Body

For Equation 3,  $M$  is 60 kg and  $H$  is 1.70 m for our experiment.

Table 3: Human Body Radiation Data

| Parameter                  | Value                     |
|----------------------------|---------------------------|
| Radius                     | $10^{-2}$ m               |
| Effective Source Area      | $3.14 \times 10^{-4} m^2$ |
| Sensor Reading             | 740 mV                    |
| $P_{measured}$             | 26.4 W                    |
| Room Temperature ( $T_2$ ) | 294 K                     |
| Hand Temperature ( $T_1$ ) | 310 K                     |

While calculation of the  $P_{measured}$ , we use sensor reading with dividing 1000 because of the settings of the sensor. Also, I used sensitivity of thermopile,

$$P_{measured} = \frac{0.74}{28} = 0.0264 \text{ W}$$

By Equation 2,

$$P_{Total_1} = P_{measured} \frac{0.202 \times 60^{0.425} \times 1.70^{0.725}}{3.14 \times 10^{-4}} = 142.2 \text{ W}$$

By Equation 4,

$$P_{Total} = 2.25 \times 10^{-4} \cdot 0.98 \cdot 5.7 \times 10^{-8} \cdot (310^4 - 294^4) \cdot \frac{0.202 \times 60^{0.425} \times 1.77^{0.725}}{3.14 \times 10^{-4}}$$

$$P_{Total_2} = 122.9 \text{ W}$$

The percentile error is

$$\%Error = \frac{|P_{Total_1} - P_{Total_2}|}{P_{Total_2}} = 15.7\%$$

## 5 Questions

### Part 1: Stefan-Boltzmann Law

1. What is the relationship between the radiation intensity and the temperature of the filament? Does it obey Stefan-Boltzmann Law?

**Answer:** The radiation intensity is proportional to the fourth power of the absolute temperature ( $T^4$ ). Our experimental data demonstrate a linear relationship in Figure 1, confirming that the filament obeys the Stefan-Boltzmann Law.

2. The Stefan-Boltzmann Law is perfectly true only for ideal, black body radiation. A black body is any object that absorbs all the radiation that strikes it. Is the filament of the lamp a true blackbody?

**Answer:** No, the filament is not a true blackbody. It is a grey body because its surface has an emissivity of less than one. While it obeys the Stefan-Boltzmann Law, it radiates less energy than an ideal blackbody.

3. What sources of thermal radiation, other than the lamp filament, might have influenced your measurements? What affect would you expect these sources to have on your results?

**Answer:** The light in the environment increases that. We also sometimes affect it, the value changes when we get closer. These may cause an increase in the values, that is why we zero the first value.

## Part 2: Human Body Power Calculation

1. According to your measurements, estimate your daily Calorie intake in order to compensate the energy loss due to heat radiation?

**Answer:** I will use the  $P_{Total} = 122.94$  W, the energy loss per second is 122.94 J. In one day, we have  $24 \times 3600 = 86400$  s, so the total energy loss is:

$$\begin{aligned} E_{day} &= 122.94 \text{ J/s} \times 86400 \text{ s} \\ &\approx 1.062 \times 10^7 \text{ J} \end{aligned}$$

Since 1 kcal = 4184 J:

$$\text{Calories} = \frac{1.062 \times 10^7 \text{ J}}{4184 \text{ J/kcal}} \approx 2538 \text{ kcal}$$

2. Why an Aluminum-styrofoam sheet is placed between the sensor and the hand in this experiment. Are there any other materials that you would use instead?

**Answer:** The Aluminum-styrofoam sheet is used to fix the source area and as an insulator. The aluminum side of the sensor reflects external radiation, while the styrofoam side prevents heat transfer with hand. Other materials with similar species can be used but other than these type materials may increase the error in the experiment.